Technology Stack

Date	15 March 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*HTML, CSS, JavaScript*	*Provides a responsive and user-friendly interface for farmers to enter agricultural data and view crop recommendations.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python (Flask)*	*Supports machine learning integration and efficient processing of soil and environmental data.*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*MySQL*	*Stores crop information, soil data, weather data, and recommendation results efficiently.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*AWS Cloud*	*Provides scalable, reliable, and cost-effective hosting for the application and database.*
5	**Version Control & CI/CD**	*GitHub*	*Enables version control, team collaboration, and project management.*

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Weather API, Scikit-Learn, Pandas, NumPy*	*Weather API provides environmental data, while Scikit-Learn, Pandas, and NumPy support machine learning and data analysis for crop prediction.*